Priority Queue

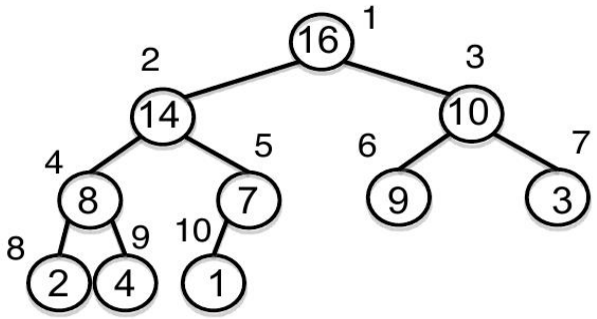
A data structure implementing a set S of elements, each associated with a key, supporting the following operations:

insert(S, x) :	insert element x into set S
max(S) :	return element of S with largest key
extract_max(S) :	return element of S with largest key and remove it from S
increase_key(S, x, k) :	increase the value of element x' s key to new value k (assumed to be as large as current value)

Heap

- Implementation of a priority queue
- An array, visualized as a nearly complete binary tree
- Max Heap Property: The key of a node is $\geq$ than the keys of its children

(Min Heap defined analogously)



1	2	3	4	5	6	7	8	9	10
16	14	10	8	7	9	3	2	4	1

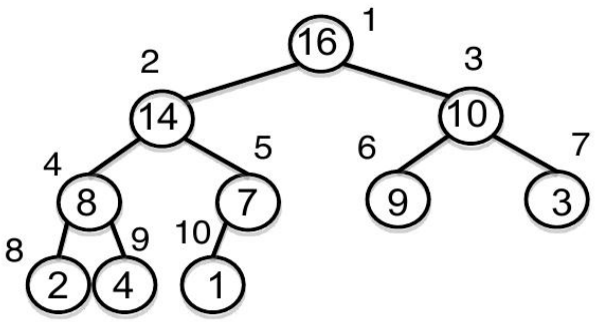
Heap as a Tree

root of tree: first element in the array, corresponding to i = 1

parent(i) =i/2: returns index of node's parent

left(i)=2i: returns index of node's left child

right(i)=2i+1: returns index of node's right child



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16	14	10	8	7	9	3	2	4	1

No pointers required! Height of a binary heap is O(lg n)

Heap Operations

build_max_heap : produce a max-heap from an unordered array

max_heapify : correct a single violation of the heap property in a subtree at its root

insert, extract_max, heapsort

Max_heapify

• Assume that the trees rooted at left(i) and right(i) are max-heaps

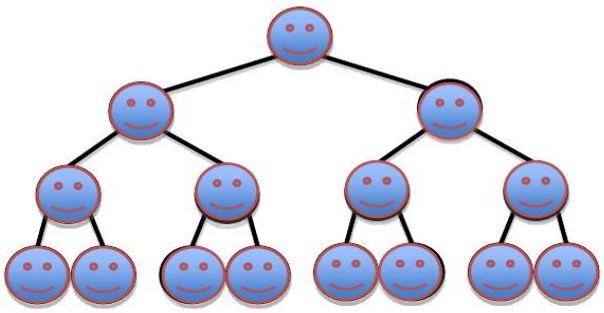
• If element A[i] violates the max-heap property, correct violation by "trickling" element A[i] down the tree, making the subtree rooted at index i a max-heap

Max_Heapify Pseudocode

```
l = left(i)
r = right(i)
if (l <= heap-size(A) and A[l] > A[i])
    then largest = l     else largest = i
if (r <= heap-size(A) and A[r] > A[largest])
    then largest = r
if largest ≠ i
    then exchange A[i] and A[largest]
            Max_Heapify(A, largest)
```

Build_Max_Heap(A)

Converts A[1…n] to a max heap



```
Build_Max_Heap(A):
    for i=n/2 downto 1
        do Max_Heapify(A, i)
```

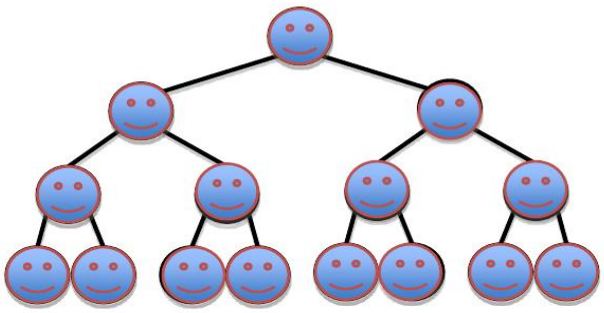
Why start at n/2?

Because elements A[n/2 + 1 … n] are all leaves of the tree
$2i > n$, for $i > n/2 + 1$

Time=? O(n log n) via simple analysis

Build_Max_Heap(A) Analysis

Converts A[1…n] to a max heap



```
Build_Max_Heap(A):
    for i=n/2 downto 1
        do Max_Heapify(A, i)
```

Observe however that Max_Heapify takes O(1) for time for nodes that are one level above the leaves, and in general, O(l) for the nodes that are l levels above the leaves. We have n/4 nodes with level 1, n/8 with level 2, and so on till we have one root node that is lg n levels above the leaves.

Build_Max_Heap(A) Analysis

Converts A[1…n] to a max heap

```
Build_Max_Heap(A):
    for i=n/2 downto 1
        do Max_Heapify(A, i)
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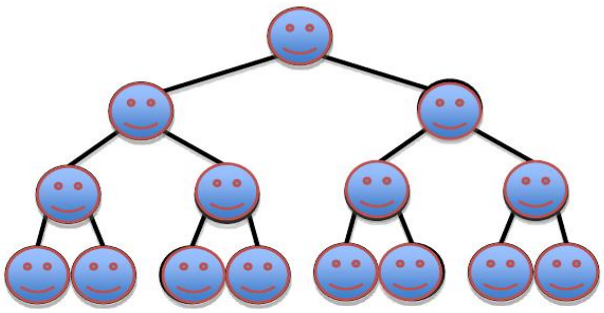

Total amount of work in the for loop can be summed as:

$n/4\ (1\ c) + n/8\ (2\ c) + n/16\ (3\ c) + \ldots + 1\ (\lg n\ c)$

Setting $n/4 = 2^k$ and simplifying we get:

$c\ 2^k(1/2^0 + 2/2^1 + 3/2^2 + \ldots (k+1)/2^k)$

The term is brackets is bounded by a constant!

This means that Build_Max_Heap is $O(n)$

Heap-Sort

Sorting Strategy:

1. Build Max Heap from unordered array;
2. Find maximum element A[1];
3. Swap elements A[n] and A[1]:
   now max element is at the end of the array!
4. Discard node n from heap
   (by decrementing heap-size variable)
5. New root may violate max heap property, but its children are max heaps. Run max_heapify to fix this.
6. Go to Step 2 unless heap is empty.

Heap-Sort

Running time:

after n iterations the Heap is empty

every iteration involves a swap and a max_heapify operation; hence it takes O(log n) time

Overall O(n log n)